JAVASCRIPT ASSIGNMENT : 13

Module 13: Scope, Hoisting and Closures – Multiple Assignment Questions

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Section 1: Scope Types

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1. Demonstrate global scope

Create variable outside function and access it inside function.
Hint: Global variable accessible everywhere.

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2. Demonstrate function scope

Create variable inside function and try accessing it outside.
Hint: It should not be accessible outside.

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3. Demonstrate block scope using let

Declare variable inside if block using let and try accessing outside block.
Hint: let is block scoped.

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4. Compare var and let scope

Create loop using var and another using let and print variable outside loop.
Hint: Observe difference.

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5. Create shadowing example

Create global variable and inside function create variable with same name. Print both values.
Hint: Inner variable shadows outer variable.

Section 2: Hoisting

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6. Demonstrate variable hoisting with var

Use variable before declaration and observe output.
Hint: var is hoisted but initialized as undefined.

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7. Demonstrate let and const hoisting behavior

Try accessing let variable before declaration.
Hint: Temporal dead zone concept.

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8. Demonstrate function declaration hoisting

Call function before defining it and observe behavior.
Hint: Function declaration is hoisted fully.

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9. Demonstrate function expression hoisting

Call function expression before declaration.
Hint: It should give error.

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10. Predict output hoisting question

Write program with mixed var and function and predict output.
Hint: Understand execution context.

Section 3: Closures

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11. Basic closure example

Create function inside function and return inner function. Access outer variable from inner function.
Hint: Inner function remembers outer scope.

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12. Create counter using closure

Create function that returns counter function which increases value every time called.
Hint: Use private variable inside outer function.

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13. Create private variable using closure

Create function where variable cannot be accessed directly but only through returned methods.
Hint: Encapsulation concept.

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14. Closure inside loop problem

Create loop with var and create functions inside loop and observe behavior.
Hint: Classic interview question.

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15. Fix closure loop issue using let

Fix above problem using let.
Hint: Block scope solves issue.

Section 4: Real World Use Cases

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16. Create simple module pattern using closure

Create function that exposes only specific methods and hides internal variables.
Hint: Return object with selected methods.

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17. Create bank account system using closure

Create account where balance is private and can only be modified using deposit and withdraw methods.
Hint: Use closure for data protection.

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18. Create once function

Create function that runs only once even if called multiple times.
Hint: Use closure to store state.

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19. Create memoization example

Create function that stores previous calculation result and returns cached value if same input given.
Hint: Use object inside closure.

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20. Create custom timer using closure

Create function that keeps internal count and increments every second.
Hint: Combine closure and timeout.

Section 5: Interview Tricky Questions

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21. Predict output for nested scope example

Create nested functions and ask which variable will be accessed.
Hint: Lexical scope concept.

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22. What is lexical scope

Write explanation with example code.
Hint: Scope determined at definition time.

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23. Can closure cause memory leak

Explain scenario where closure keeps reference to outer variables.
Hint: Garbage collection concept.

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24. Explain difference between scope and closure

Write explanation with example.
Hint: Scope defines access, closure preserves it.

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25. Create IIFE with closure example

Create immediately invoked function and demonstrate private variable.
Hint: Use function wrapping.

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